

INVESTIGATION / INQUIRY ACTIVITY

Project Evidence Investigation Lab

What Do We Know, and What Must We Find Out?

Duration: 25 minutes | Work: individual then pairs | Output: evidence-based investigation brief

Investigation mission. The sponsor describes the Smart Campus project as “mostly on track,” but the available evidence is incomplete and sometimes contradictory. Investigate the case before recommending action. Separate facts from interpretations, identify missing information, and formulate questions that would improve the PMO decision.

1. Learning Outcomes

Students will be able to locate relevant evidence, distinguish facts from assumptions, connect stakeholder needs to project constraints, identify information gaps, and formulate focused inquiry questions.

2. Case Evidence File

Item	Evidence available at the start of the investigation
E1	The approved project deadline is 12 weeks and the budget is USD 40,000.
E2	The sponsor requires a demonstration at the end of week 6.
E3	At week 6, planned progress is 50% and reported actual progress is 38%.
E4	Planned cost at week 6 is USD 20,000; actual cost is USD 24,000.
E5	Two of four planned modules are complete, and 12 critical defects remain open.
E6	Database integration is identified as the principal technical risk.
E7	Developers report overload and repeated rework, but no workload records are supplied.
E8	The sponsor says all proposed features are “important,” but no formal priority list exists.
E9	Students need secure login, course schedules, and notifications for the pilot.
E10	No current evidence describes data privacy approval, acceptance-test results, or user feedback.

3. Fact, Interpretation, or Assumption? (5 minutes)

Classify each statement and cite the evidence item that supports or challenges it.

Statement	Fact	Interpretation	Assumption	Evidence ID
The project is behind schedule.	☐	☐	☐	
The development team needs more staff.	☐	☐	☐	
The pilot cannot succeed in week 12.	☐	☐	☐	
Database integration threatens essential features.	☐	☐	☐	
All proposed features must be in the first pilot.	☐	☐	☐	
The project has a quality problem.	☐	☐	☐	

4. Stakeholder Evidence Map (5 minutes)

For each stakeholder, identify a likely need and the evidence still required. Do not invent answers.

Stakeholder	Need or concern supported by the case	Missing evidence to investigate
Students		
University sponsor		
IT and database services		
Development team		
Data-protection / security staff		

5. Constraint and Risk Investigation (5 minutes)

Most urgent confirmed constraint _____

Most serious confirmed risk _____

Important uncertainty _____

Evidence connecting them _____

Possible cause chain:

Evidence → possible cause → project effect → stakeholder consequence

Complete one chain:

6. Build High-Value Inquiry Questions (5 minutes)

A useful inquiry question is specific, answerable with evidence, and capable of changing a decision. Write one question for each category.

Category	Investigation question
Scope priority	
Schedule forecast	
Integration cause	
Quality readiness	
Team capacity	
Stakeholder acceptance	

Circle the question that would have the greatest effect on the recovery decision. Explain why:

7. Pair Evidence Review (3 minutes)

Exchange investigations with a partner. Your partner must identify:

- one conclusion that is strongly supported;
- one assumption that needs to be labeled; and
- one additional evidence source that should be requested.

Partner feedback:

8. Investigation Brief (2 minutes)

Complete the PMO summary:

The evidence currently shows _____. We should not yet conclude _____ because _____. Before deciding, the PMO should obtain _____.

9. Assessment and AI Boundary

Assessment: evidence classification (5), stakeholder reasoning (5), inquiry-question quality (5), and investigation brief (5), for a total of 20 points.

If AI is permitted, use it only to propose additional questions after completing your own evidence map. Verify that every AI suggestion is relevant to the supplied case; do not allow AI to invent missing facts.

Course portal: maoude.github.io/teaching-ai-pmo/